



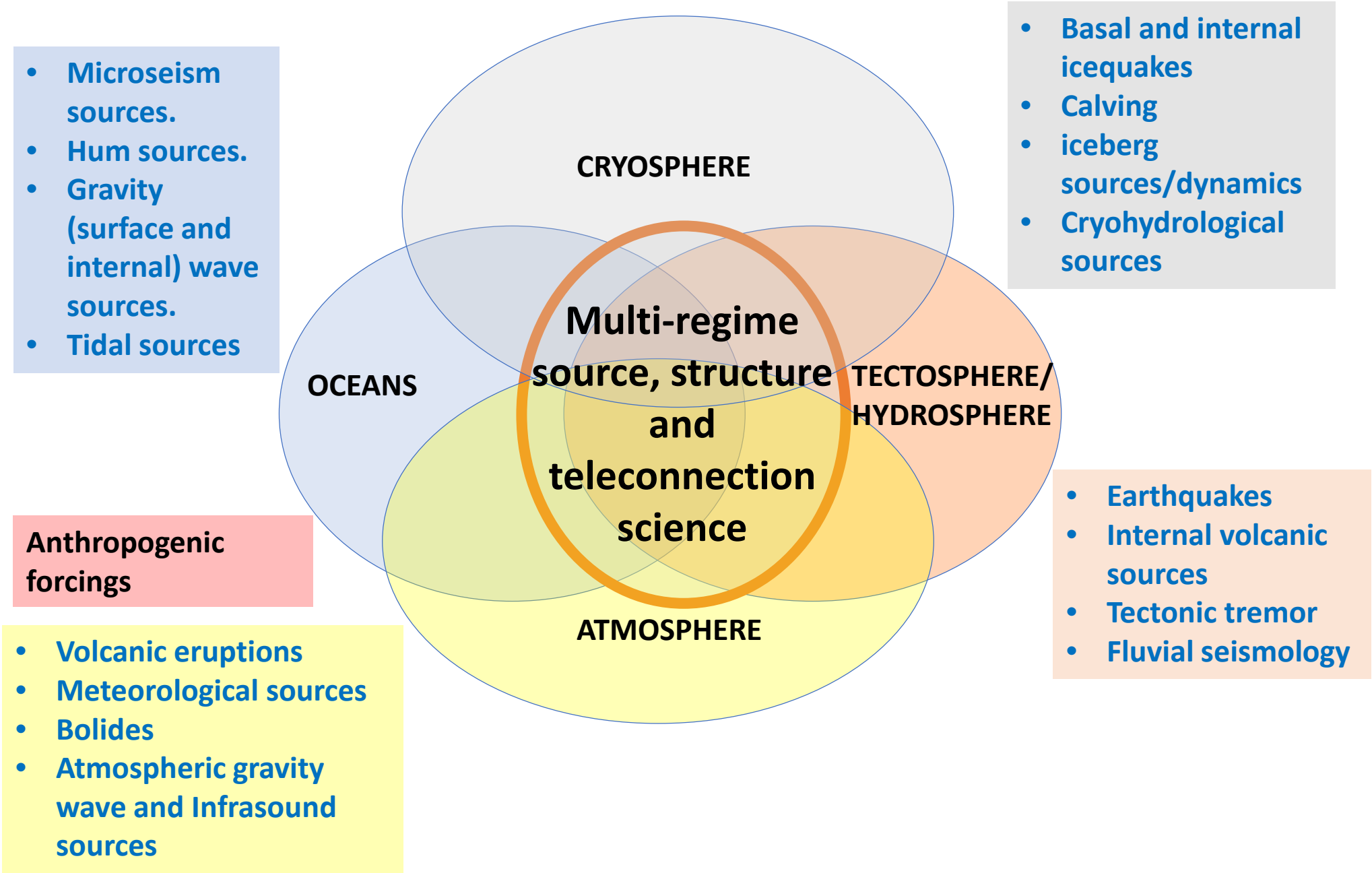
Glacial Seismology: A Rapidly Evolving Discipline

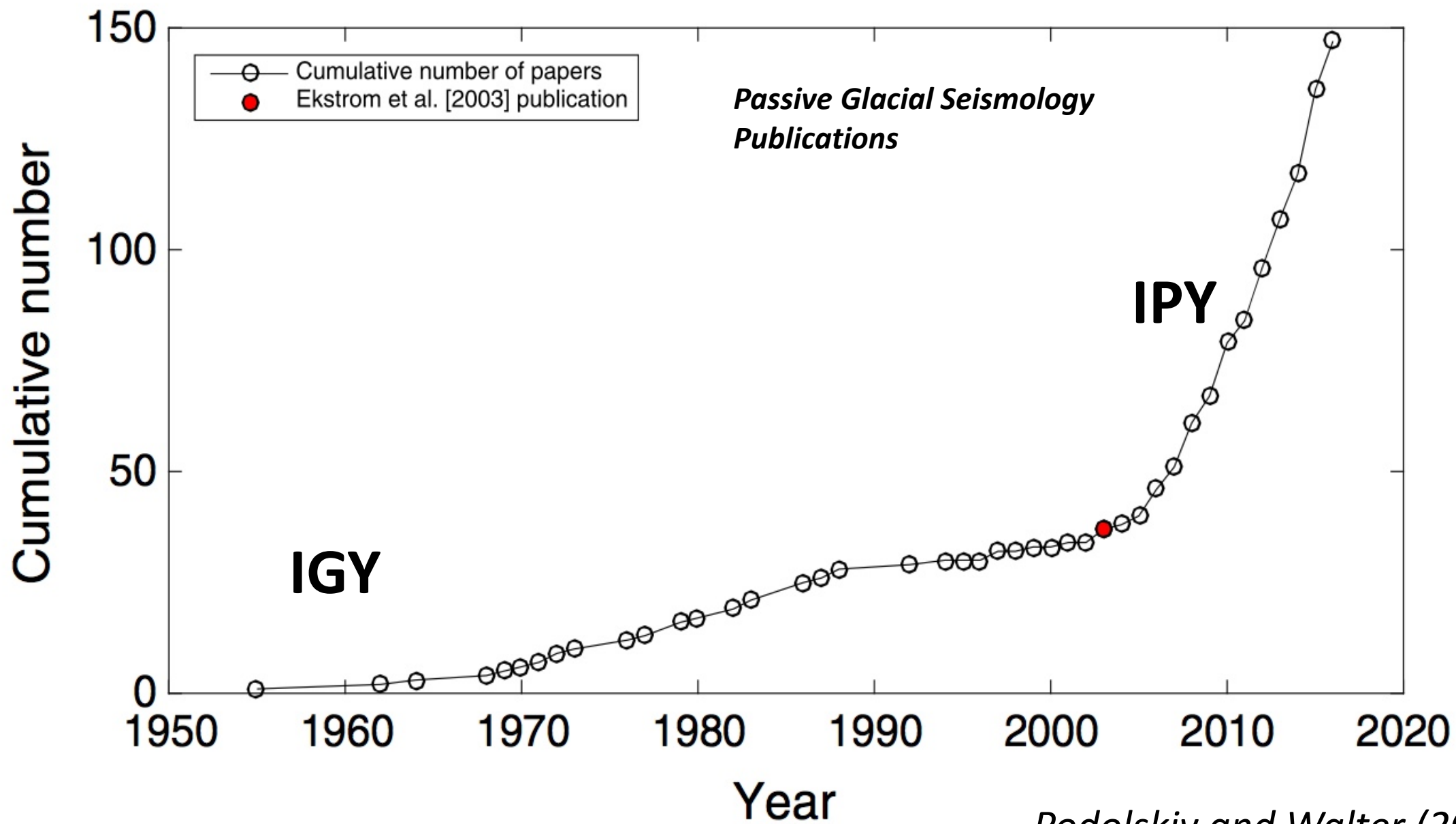
Rick Aster

**Department of Geosciences
Colorado State University**

*See posted review papers:
Podolskiy and Walter (2016)
Aster and Winberry (2017)*

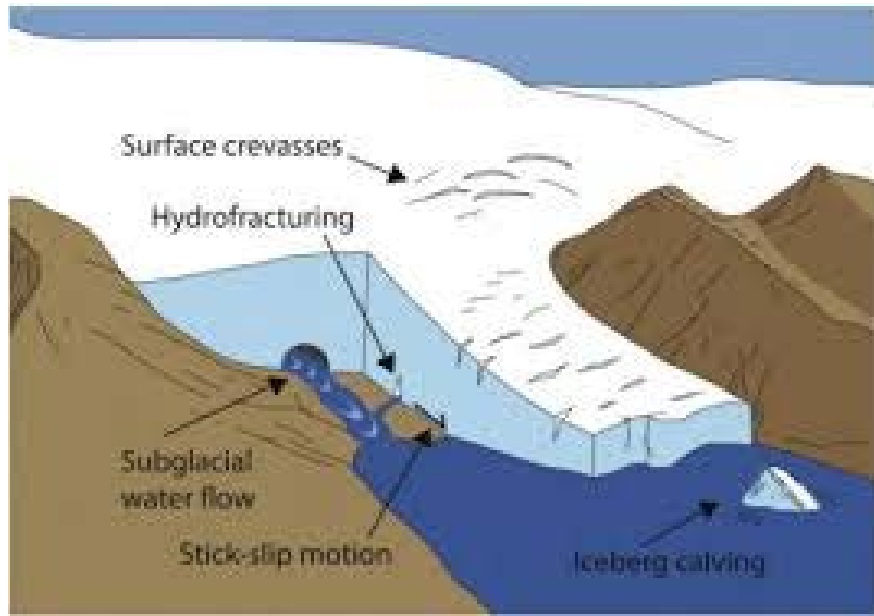
Global to local continuous seismic data now facilitate the study of a huge range of natural elastic and gravity wave sources, as well as propagation, temporal variation, and interlinked phenomena across Earth systems.



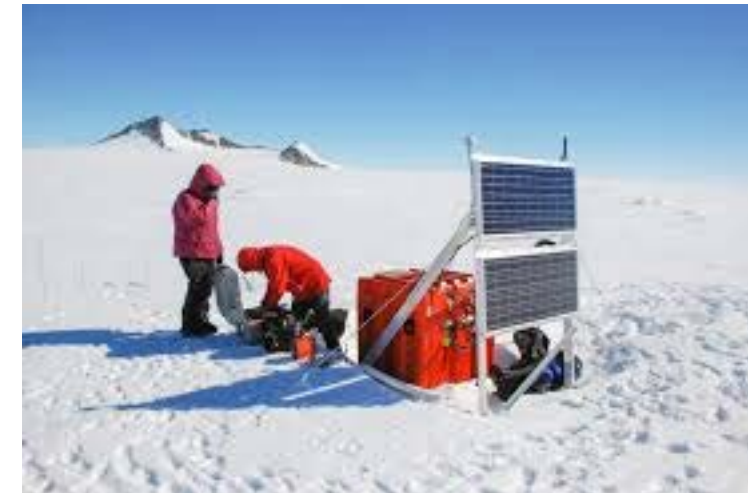


Podolskiy and Walter (2016)

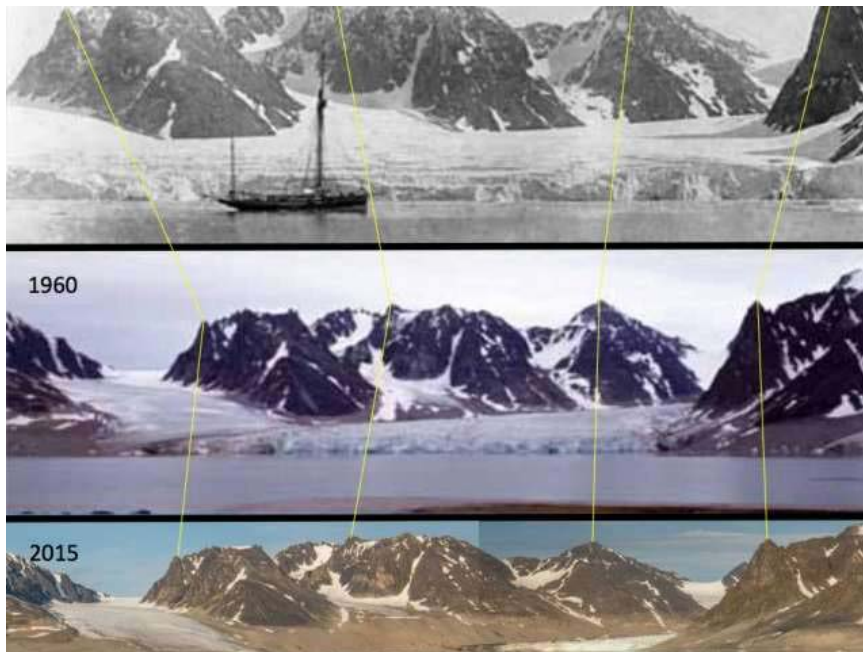
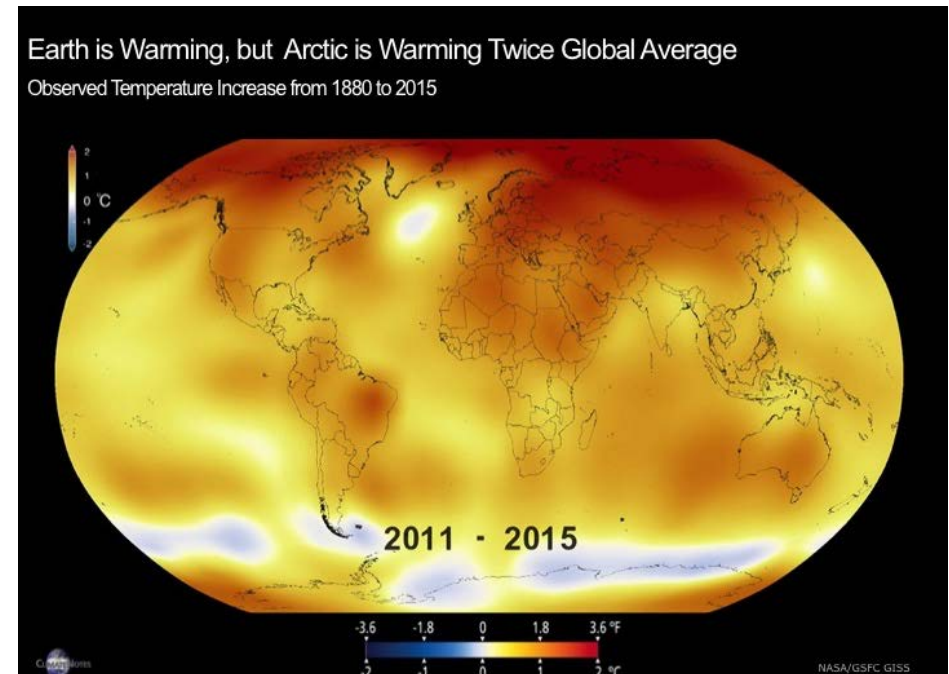
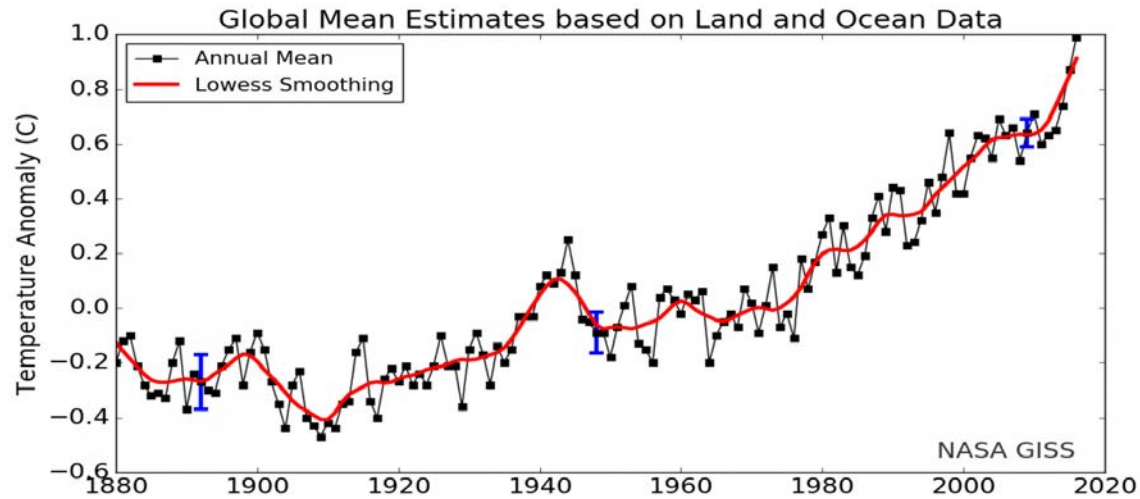
Why Glaciology (and Glacial Seismology)?



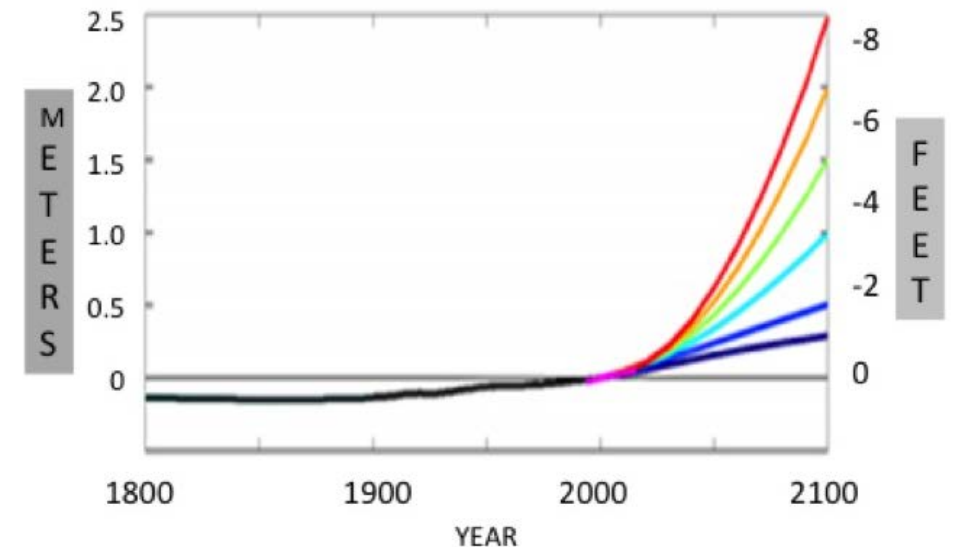
Podolskiy and Walter (2016)



A Changing Planet

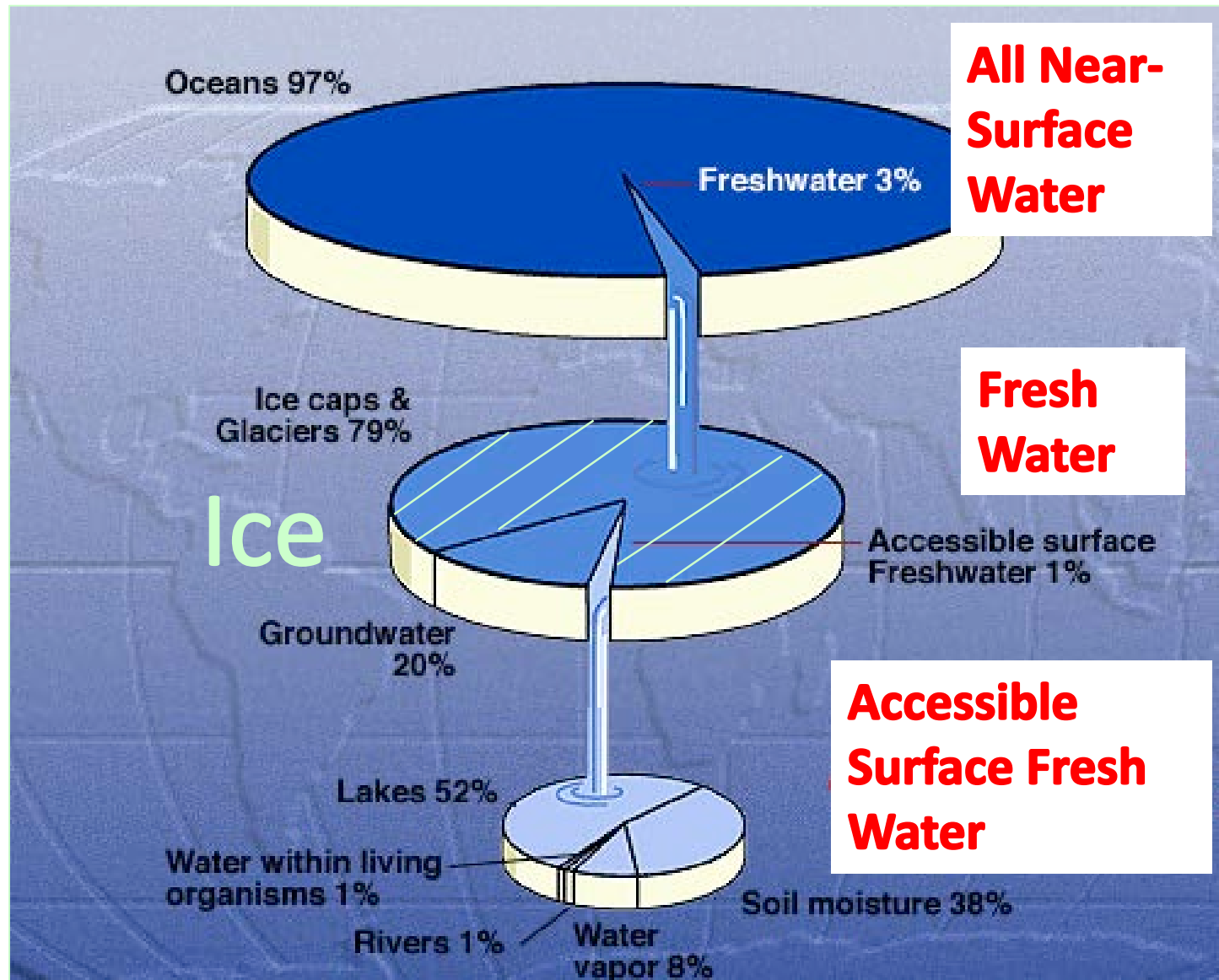


Six Projections for Rising Sea Level
2017 NOAA Technical Paper 083

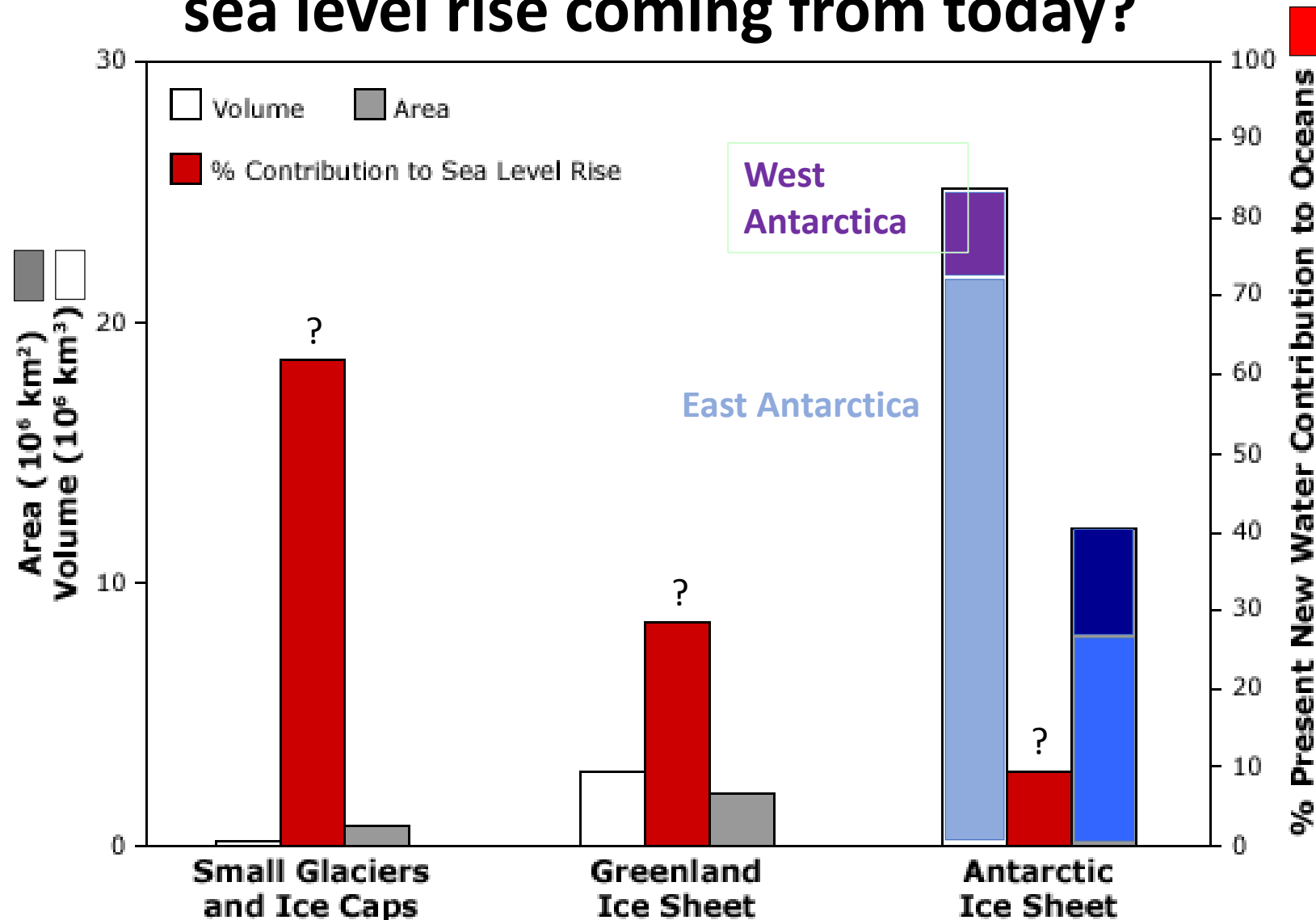


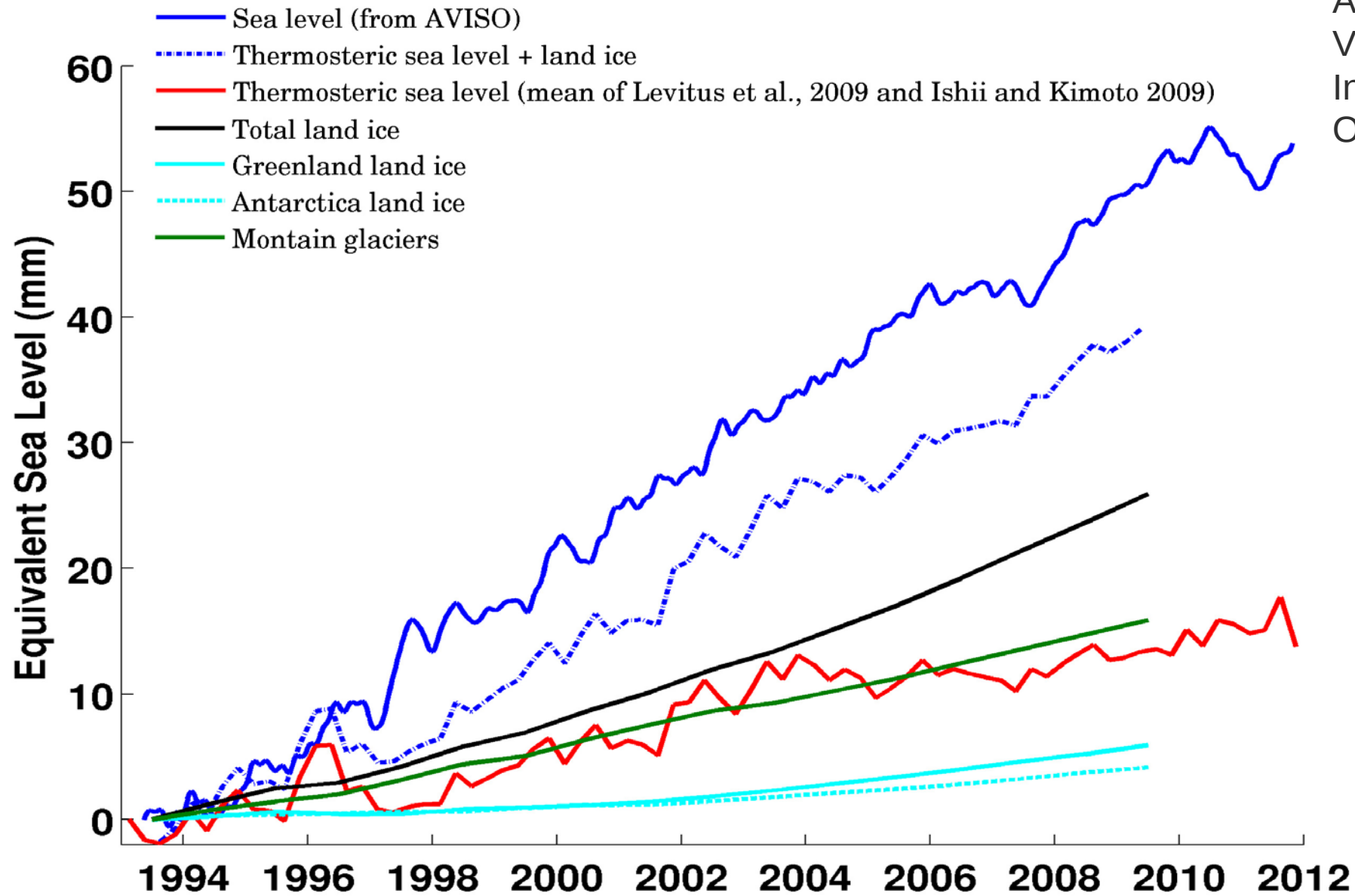
Andreas Weith

Where is Earth's Water?



Where's the ice, and where is are contributions to sea level rise coming from today?



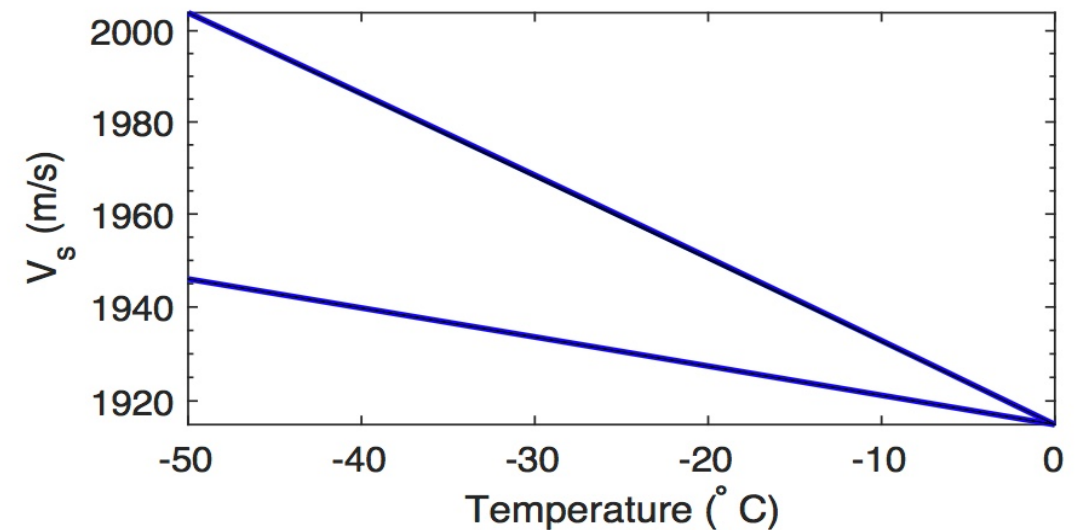
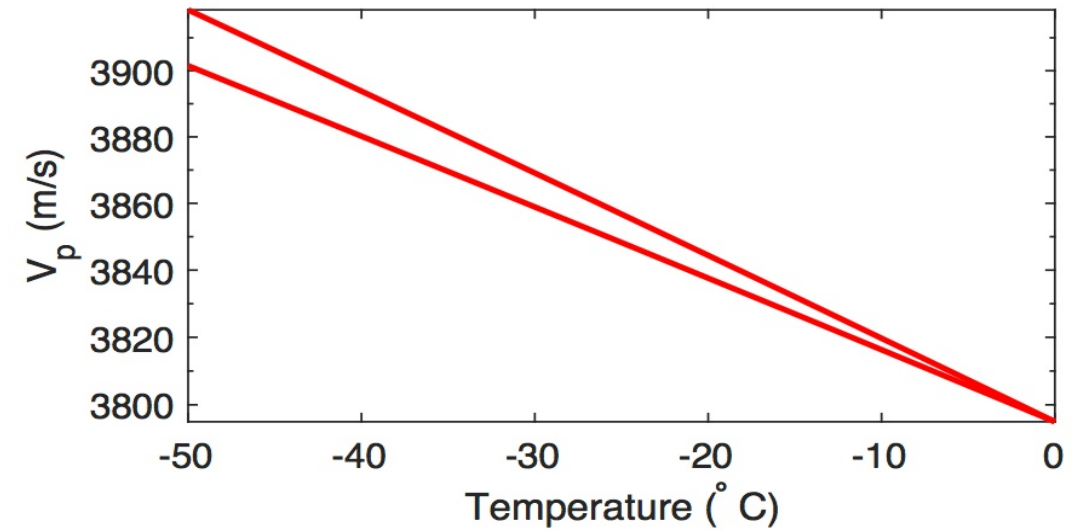


AVISO: Archiving,
Validation and
Interpretation of Satellite
Oceanographic data

*Levitus et al., 2009; Ishii
and Kimoto, 2009; NASA*

Seismology: the *Study of Elastic Wave* $\vec{f} = m\vec{a}$ and Their Sources

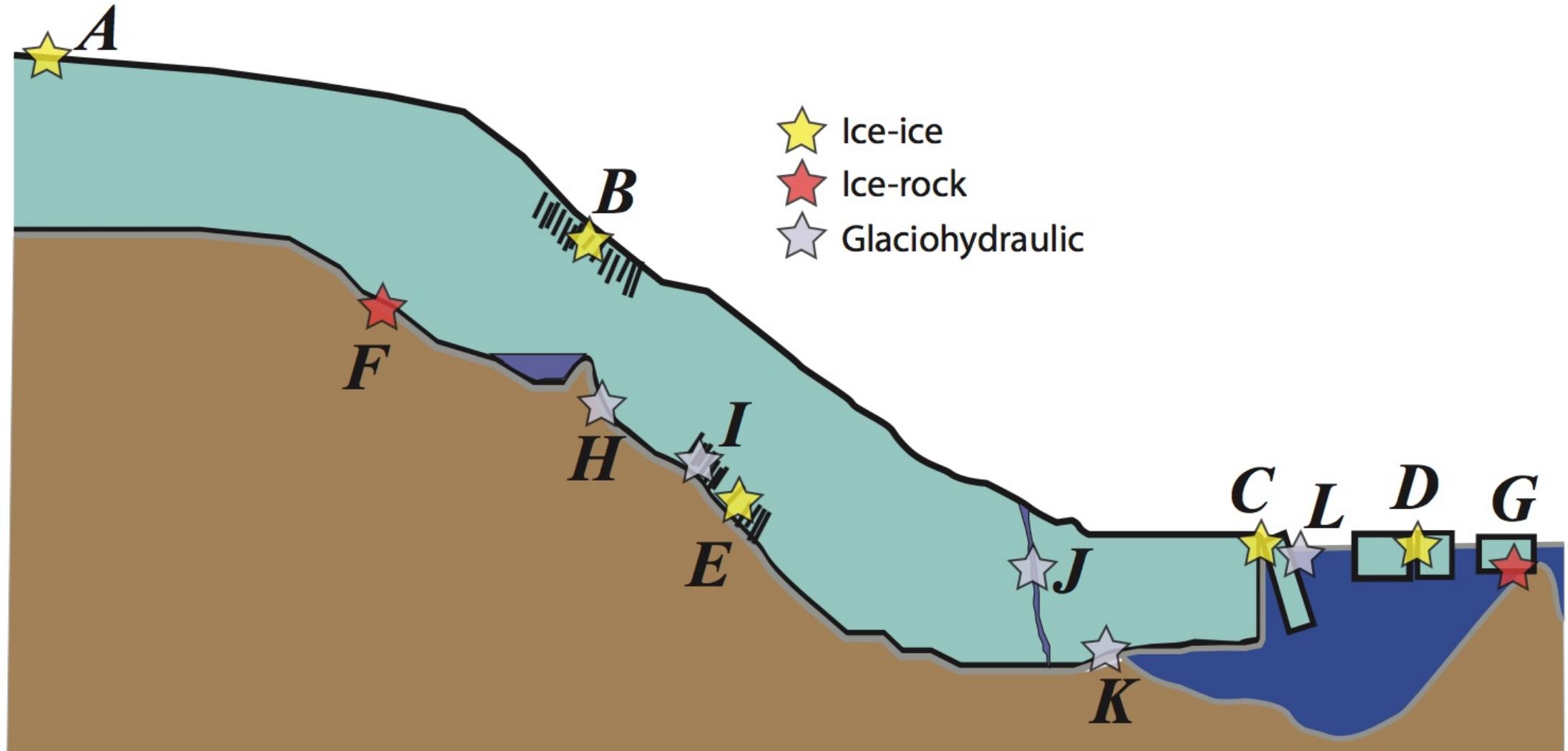
- Ice deforms viscously at slow strain rates, but behaves elastically at rapid strain rates (that generally result in very low absolute strains), thus propagating (generally low attenuation) seismic waves.
- Elastic wave speeds in Ice are comparable to crustal rocks near Earth's surface, so elastic wavefields between ice and rock are commonly well coupled (not so much with water or air).



Seismology (a broader definition): *Study of Elastic Wave and Their Sources, and:*

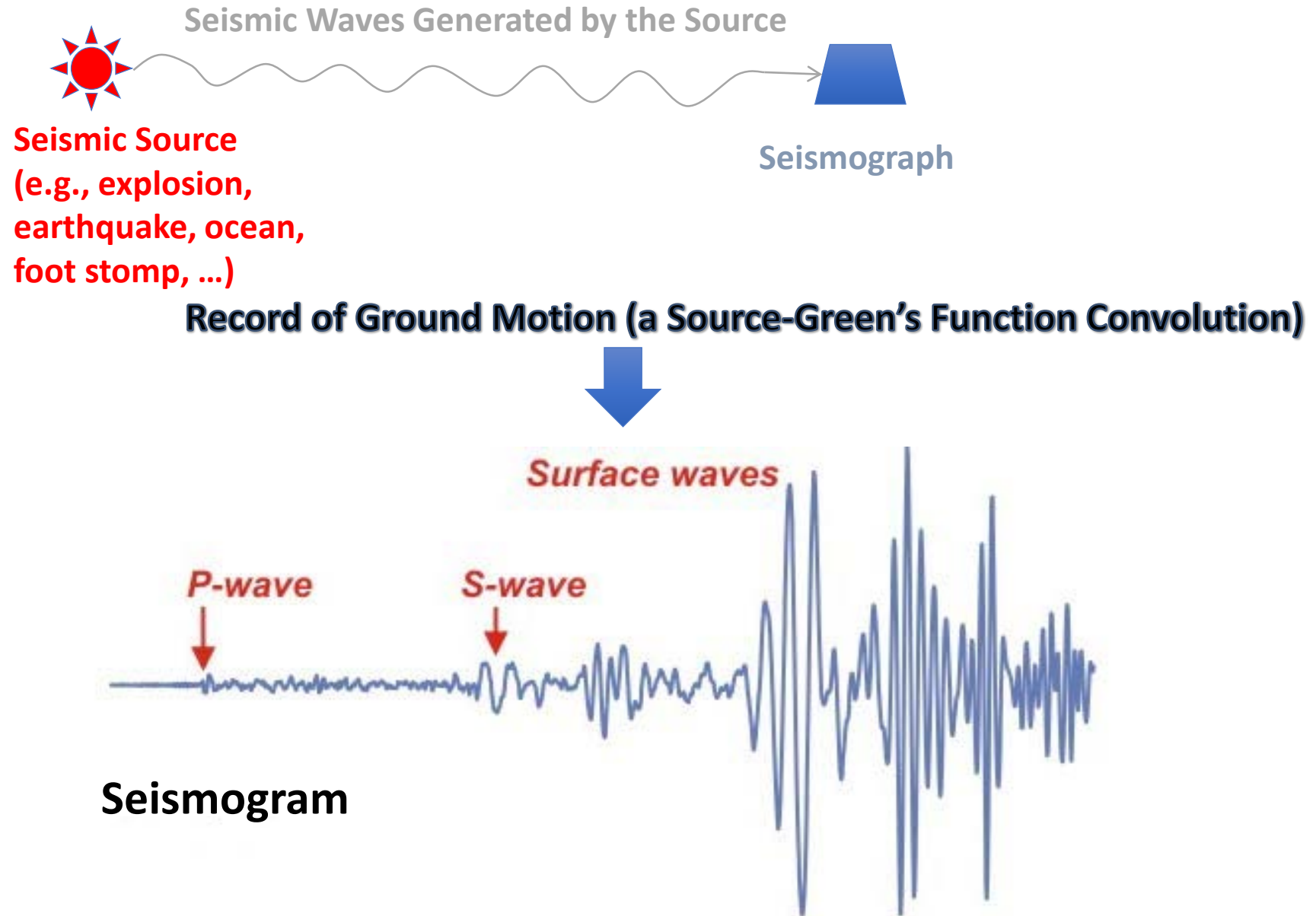
- Utilization of seismic waves for delineating elastic and elastic attenuation features and structure.
- Coupling of the elastic regime with acoustic and gravity wave systems.
- Interpretation of seismic source forces with the diverse transient processes by which they are generated (fault slip, magmatic, explosive, fluid flow, ...)

A Plethora of Cryoseismic Source Processes!



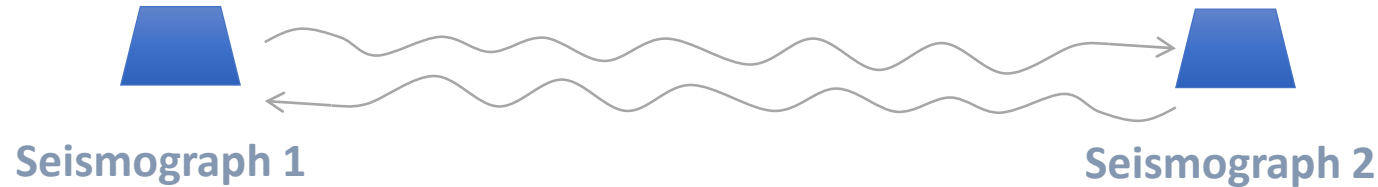
Aster and Winberry (2017)

“Conventional” Seismology

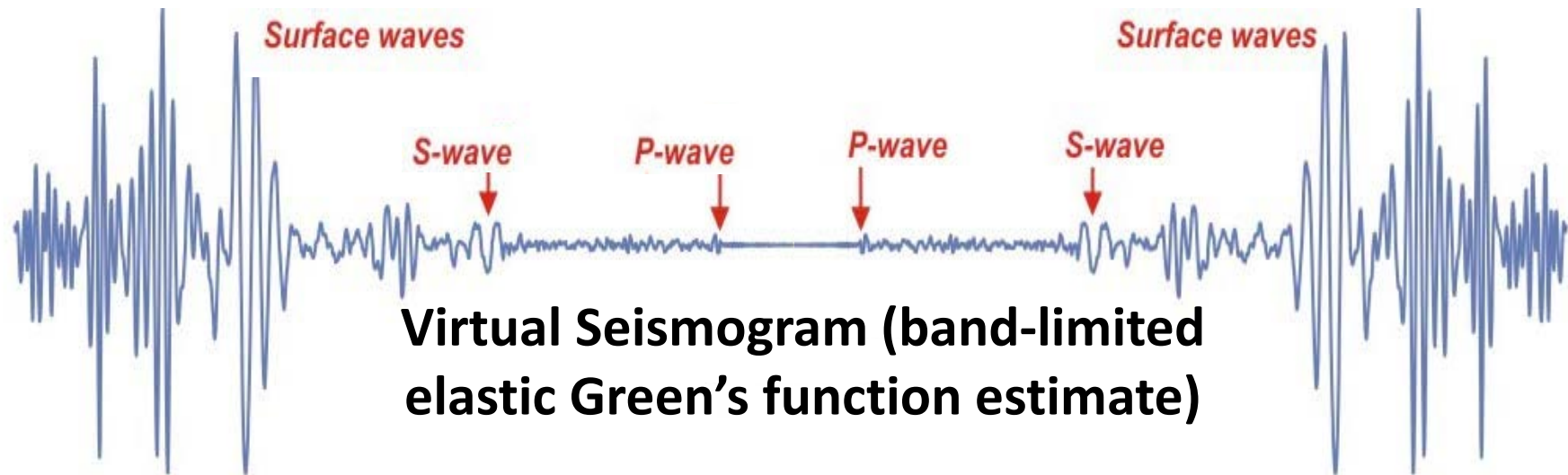


Ambient Noise Seismology (a seismological revolution)

Faint Seismic Waves Generated by Wind, Waves,
and Other Background “Noise” Processes



Correlation stack from (very long) Records of Ground Motion



**Virtual Seismogram (band-limited
elastic Green's function estimate)**

Seismic waves traveling from
stations 2 -> 1 (time-reversed)

Seismic waves traveling from
station 1 -> 2

Glacial Seismology has substantially adapted and evolved methods that were largely (although not exclusively) initially developed in other contexts. Some notable examples include:

- Tremor/continuous signal and source studies (tectonic, volcanic, microseism, ...)
 - e.g., Iceberg Harmonic Tremor; MacAyeal, D. R., E. A. Okal, R. C. Aster, and J. N. Bassis (2008), Seismic and hydroacoustic tremor generated by colliding icebergs, *Journal of Geophysical Research-Earth Surface*, 113(F3), doi:10.1029/2008JF001005.
- Correlation-based Green's function estimation methods (tectonic and volcanic)
 - e.g., Greenland seasonal tomography; Mordret, A., T. D. Mikesell, C. Harig, B. Lipovsky, and G. A. Prieto (2016), Monitoring southwest Greenland's ice sheet melt with ambient seismic noise, *Science Adv.*, 2, doi:10.1126/sciadv.1501538.
- Source inversion (for equivalent body forces and force couples, and their temporal evolution).
 - e.g., Slow ice-stream events; Wiens, D. A., S. Anandakrishnan, J. P. Winberry, and M. A. King (2008), Simultaneous teleseismic and geodetic observations of the stick-slip motion of an Antarctic ice stream, *Nature*, 453(7196), 770-773.

Glacial Seismology has substantially adapted and evolved methods that were largely (although not exclusively) initially developed in other contexts. Some notable examples include:

- High-precision source location and characterization
 - e.g., seismogenic crevasse formation; Walter, F., J. F. Clinton, N. Deichmann, D. S. Dreger, S. E. Minson, and M. Funk (2009), Moment tensor inversions of icequakes on Gornergletscher, Switzerland, *Bull. Seismol. Soc. Am.*, 99(2A), 852–870, doi:10.1785/0120080110.
- Back projection, gradiometry, and other “full wavefield” analyses facilitated by large seismographic networks/arrays.
 - Back projection has been used at the global scale to study large calving-associated glacial earthquakes; Larmat, C., J. Tromp, Q. Liu, and J.-P. Montagner (2008), Time reversal location of glacial earthquakes, *Journal of Geophysical Research: Solid Earth*, 113(B9), doi:10.1029/2008JB005607
 - Back projection of autocorrelograms has also been applied using dense seismographic arrays on glaciated volcanoes; Chaput, J., M. Campillo, R. Aster, P. Roux, P. Kyle, H. Knox, and P. Czoski (2015), Multiple scattering from icequakes at Erebus volcano, Antarctica: Implications for imaging at glaciated volcanoes, *J. Geophys. Res.*, 120, doi:10.1002/2014JB011278.
 - However, applying recently achievable “large N” seismographic deployment capabilities to glaciers is still a nascent opportunity (Lemon Creek Glacier experiment occurring this season; could be opportunities for a future “community experiment”).

Improved Instrumentation and Facilities

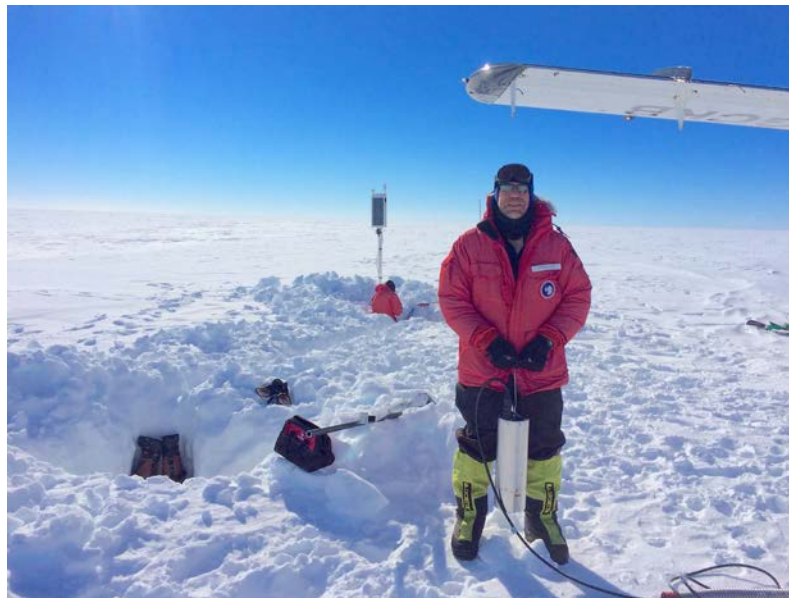


A Facility Plan for Polar Seismic and Geodetic Science: Meeting Community Needs Through IRIS and UNAVCO Polar Services

This plan has been coordinated by the Polar Networks Science Committee, a joint IRIS/UNAVCO committee, with input from a broad cross-section of the polar seismic and geodetic science community.

NOVEMBER 2012

The Facility Plan Writing Committee: Andrew Nyblade (Chair), Jason Amundson, Samantha Hansen, Erik Ivins, Matt Lazzara, Meredith Nettles, Carol Raymond, Leigh Stearns



Nyblade et al., 2012

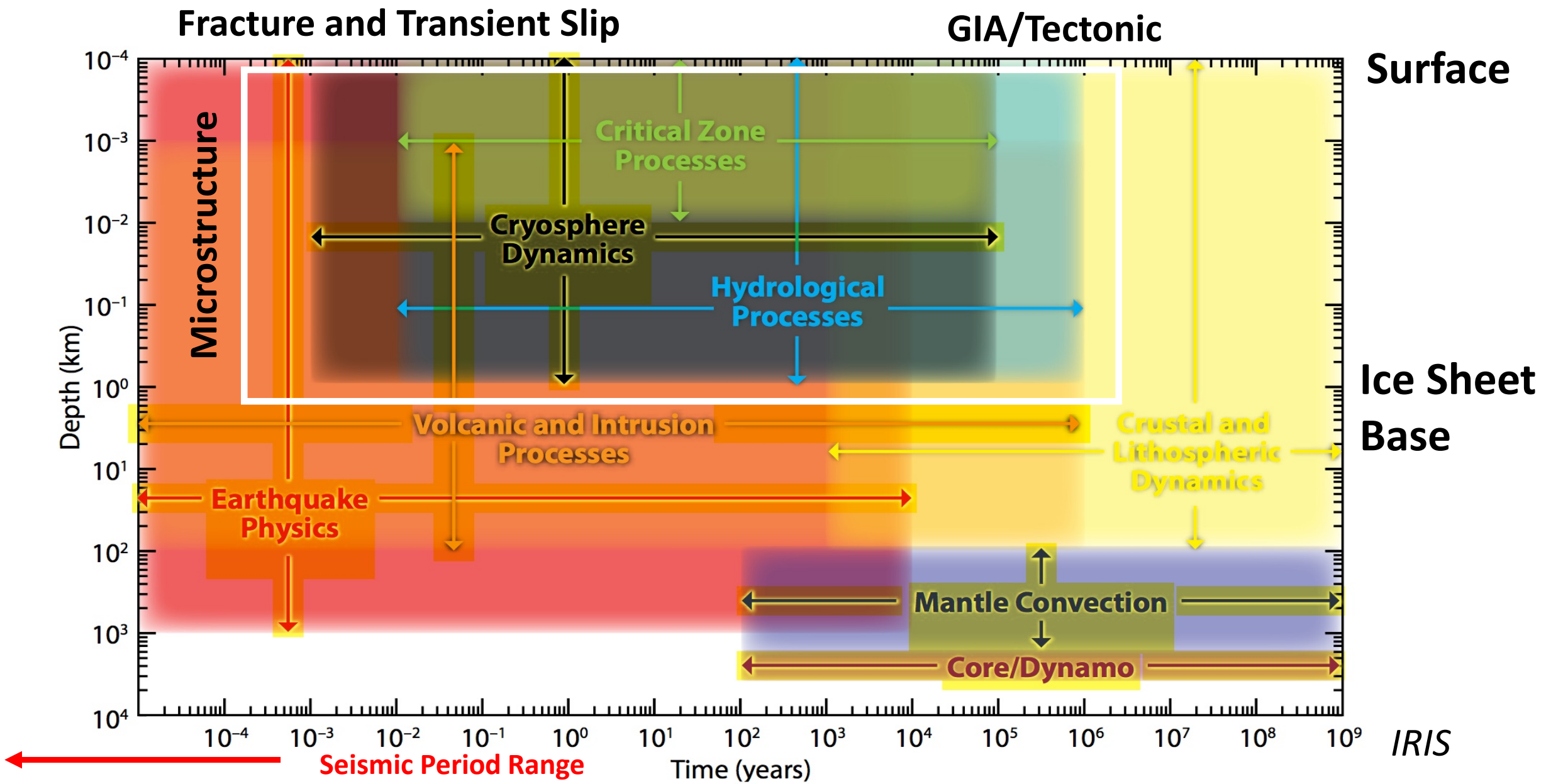
Quantification and Understanding of Dynamic Processes Producing Seismically Observable Signals

- **Basal slip**
 - “Slow” with embedded asperities
 - Repeated short-period Stick-slip
- **Internal deformation and fracture**
 - Crevassing
 - Stability of ice shelves
- **Glacial/tidewater, glacial/glacial lake, sea/lake ice, and iceberg systems**
 - Calving (large- and small-scale)
 - Iceberg collision, break-up, and tremor
 - Sea and lake ice
- **Coupling with oceanic and Solid Earth Elastic and/or Gravity Wave Systems**
 - Tides
 - Gravity waves (Infragravity <-> Swell <-> Short-period)
 - Flexural waves
 - Teleseismic, regional and local tectonic seismicity
- **Glacial hydrology**
 - Surface/near-surface
 - Englacial
 - Subglacial

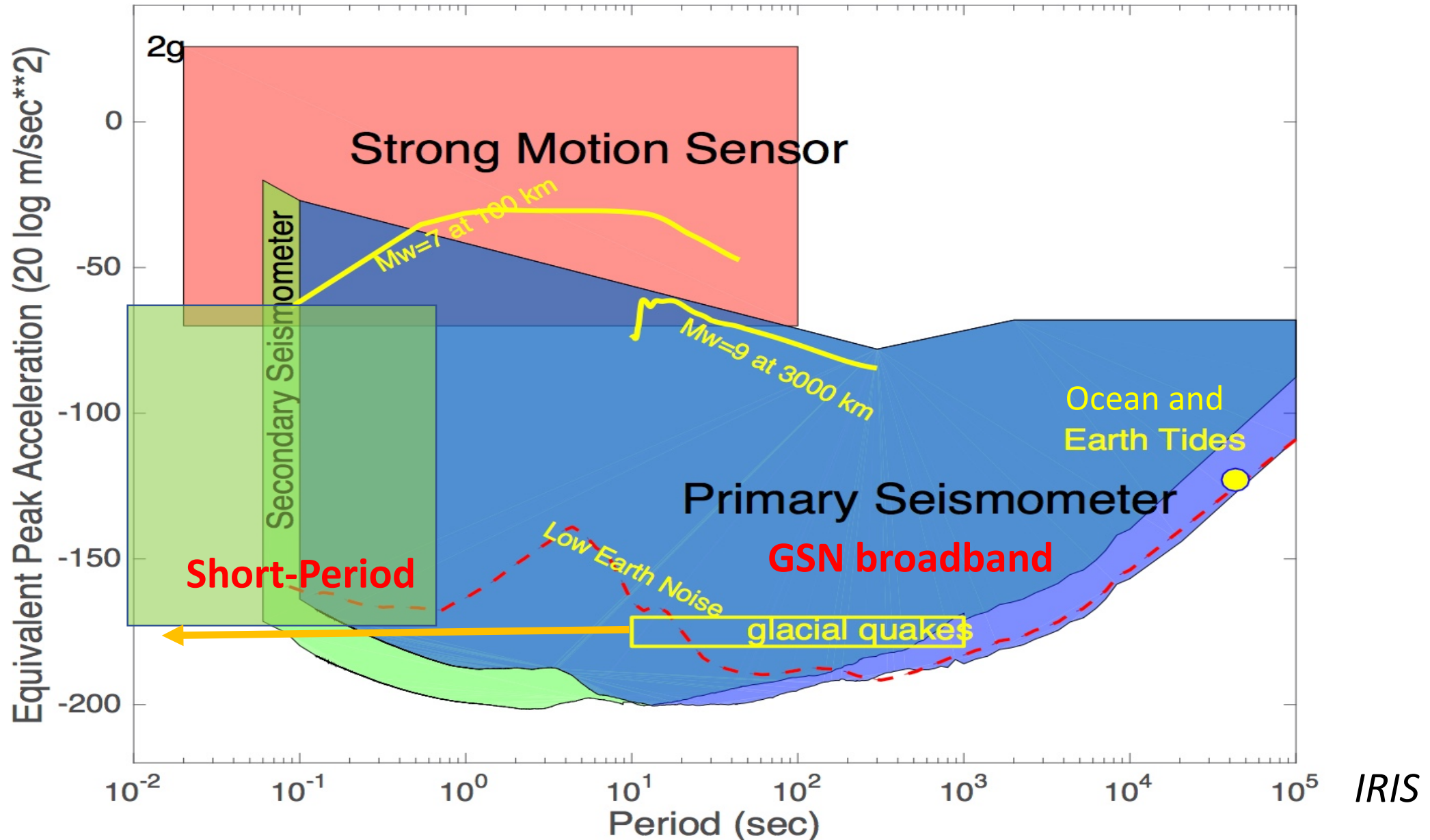
Conditions and processes; interrogated via seismic tomography and other inversion methods, and/or via time-lapse seismic measurements

- **Glacial state**
 - Temperature
 - Melt
 - Crevassing/Fracture
 - Hydrological
- **Bed Conditions**
 - Bed interface
 - Bed geology and geophysics
- **Subglacial Solid-Earth Influences**
 - Glacial isostatic adjustment processes (elastic and anelastic)
 - Heat flux
 - Tectonic history and present structure

Cryospheric Processes -- Depth and Period Regime



Seismograph Dynamic Range and Period



Earth's Seismic Spectrum

Five years of 1-hour power spectral density calculations from station QSPA (V), 7.8 km from the South Pole.

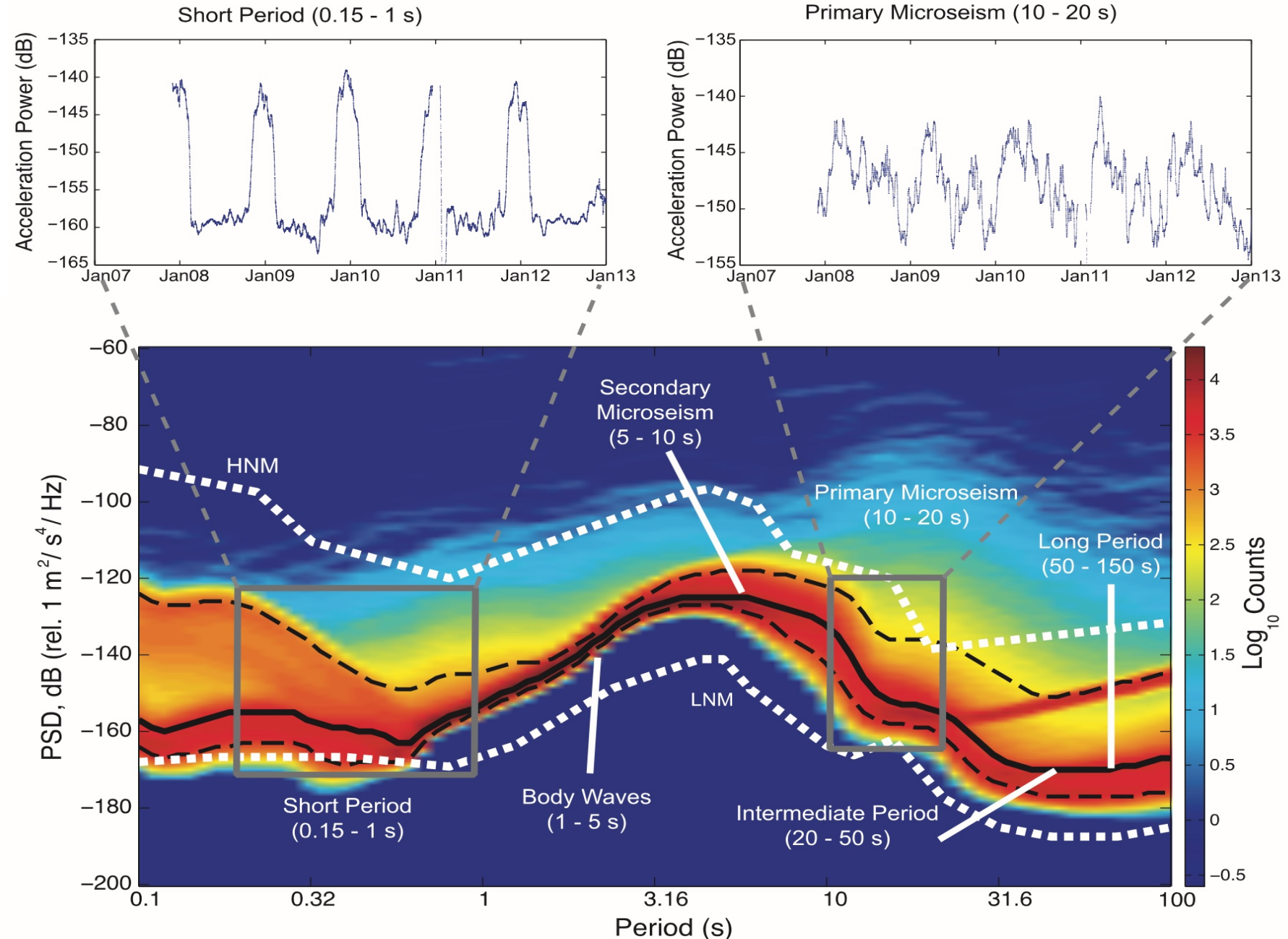
Noise between ~ 7 and ~ 30 s is dominated by the oceanic microseism.

Short period noise is most influenced by water, wind, and anthropogenic processes.

Earthquake signals in most locales are highly intermittent.

Anthony et al. (2015)

Annual Variability



Multidisciplinary Collaboration Opportunities for Glacial Seismologists

- Glaciology (in all of its diverse subfields)
- GPS/GNSS Geodesy
- Remote Sensing (photogrammetry, visible and rest of EM spectrum, gravity, ...)
- Electromagnetic Geophysical Methods (notably including ice-penetrating radar)
- Hydrology
- Oceanography
- Atmospheric Sciences
- Climatology
- GIA, Ice History, and other Climate Modeling
- Hazard and Resource Studies
- Exoseismology and Planetology
- And more...

Grand Multidisciplinary Questions for Science and Society

- **Past, Present, and Future of Glaciation**
- **Hazards**
 - Glacial outburst flooding
 - Glacial and critical zone mass wasting in deglaciated terrains
- **Climate, Ecology, and Resources**
 - Glacial-climate feedbacks
 - Atmospheric
 - Oceanic
 - Sea level rise
 - Marine terminating Glacier/Ice Sheet Dynamics and Stability (e.g., www.rollingstone.com/politics/features/the-doomsday-glacier-w481260)
 - Ice Shelf Dynamics and Stability
 - Water resources and quality
 - Ecosystem impacts of glacial retreat



May 2017

Questions & Comments

